

[STATS 531] HW5

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5.1

By the chain rule of probability, the joint density factors as

$$f_{X_{0:N}}(x_{0:N}) = f_{X_0}(x_0) \prod_{n=1}^N f_{X_n|X_0, \dots, X_{n-1}}(x_n | x_0, \dots, x_{n-1})$$

Since (X_n) is a Markov process, the future is conditionally independent of the past given the present:

$$f_{X_n|X_0, \dots, X_{n-1}}(x_n | x_0, \dots, x_{n-1}) = f_{X_n|X_{n-1}}(x_n | x_{n-1}), \quad n = 1, \dots, N$$

Therefore, we derive the joint distribution as

$$f_{X_{0:N}}(x_{0:N}) = f_{X_0}(x_0) \prod_{n=1}^N f_{X_n|X_{n-1}}(x_n | x_{n-1}) \quad \blacksquare$$

5.2

Applying the marginalization identity $f_{X|Y} = \int f_{X,Z|Y} dz$ with $X = X_n$, $Z = X_{n-1}$, $Y = Y_{1:n-1}$, we have

$$\begin{aligned} f_{X_n|Y_{1:n-1}}(x_n | y_{1:n-1}) &= \int f_{X_n, X_{n-1}|Y_{1:n-1}}(x_n, x_{n-1} | y_{1:n-1}) dx_{n-1} \\ &= \int f_{X_n|X_{n-1}, Y_{1:n-1}}(x_n | x_{n-1}, y_{1:n-1}) f_{X_{n-1}|Y_{1:n-1}}(x_{n-1} | y_{1:n-1}) dx_{n-1} \end{aligned}$$

Since (X_n) is Markov and Y_k depends only on X_k , we have $X_n \perp Y_{1:n-1} | X_{n-1}$,

$$f_{X_n|X_{n-1}, Y_{1:n-1}}(x_n | x_{n-1}, y_{1:n-1}) = f_{X_n|X_{n-1}}(x_n | x_{n-1}).$$

Thus we derive the prediction formula

$$f_{X_n|Y_{1:n-1}}(x_n | y_{1:n-1}) = \int f_{X_{n-1}|Y_{1:n-1}}(x_{n-1} | y_{1:n-1}) f_{X_n|X_{n-1}}(x_n | x_{n-1}) dx_{n-1} \quad \blacksquare$$

5.3

From the definition of conditional density and the product rule $f_{X,Y|Z} = f_{Y|X,Z} f_{X|Z}$, we can write

$$\begin{aligned} f_{X_n|Y_{1:n}}(x_n | y_{1:n}) &= \frac{f_{X_n, Y_n|Y_{1:n-1}}(x_n, y_n | y_{1:n-1})}{f_{Y_n|Y_{1:n-1}}(y_n | y_{1:n-1})} \\ &= \frac{f_{Y_n|X_n, Y_{1:n-1}}(y_n | x_n, y_{1:n-1}) f_{X_n|Y_{1:n-1}}(x_n | y_{1:n-1})}{f_{Y_n|Y_{1:n-1}}(y_n | y_{1:n-1})} \end{aligned}$$

In a POMP model the measurement Y_n depends on the latent state only through X_n , so $Y_n \perp Y_{1:n-1} | X_n$. Therefore:

$$f_{X_n|Y_{1:n}}(x_n | y_{1:n}) = \frac{f_{X_n|Y_{1:n-1}}(x_n | y_{1:n-1}) f_{Y_n|X_n}(y_n | x_n)}{f_{Y_n|Y_{1:n-1}}(y_n | y_{1:n-1})} \quad \blacksquare$$

This is the filtering formula.

To compute the conditional likelihood formula, we apply the marginalization identity $f_{X|Y} = \int f_{X,Z|Y} dz$ with $X = Y_n$, $Z = X_n$, $Y = Y_{1:n-1}$:

$$\begin{aligned} f_{Y_n|Y_{1:n-1}}(y_n | y_{1:n-1}) &= \int f_{Y_n, X_n|Y_{1:n-1}}(y_n, x_n | y_{1:n-1}) dx_n \\ &= \int f_{Y_n|X_n, Y_{1:n-1}}(y_n | x_n, y_{1:n-1}) f_{X_n|Y_{1:n-1}}(x_n | y_{1:n-1}) dx_n \\ &= \int f_{X_n|Y_{1:n-1}}(x_n | y_{1:n-1}) f_{Y_n|X_n}(y_n | x_n) dx_n \quad \blacksquare \end{aligned}$$

The last equation is implied from the POMP measurement model.

5.4

Write $Y_{n:N} = (Y_n, Y_{n+1:N})$ and apply the product rule $f_{X,Y|Z} = f_{X|Z} f_{Y|X,Z}$, we have

$$f_{Y_{n:N}|X_n}(y_{n:N} | x_n) = f_{Y_n|X_n}(y_n | x_n) f_{Y_{n+1:N}|X_n, Y_n}(y_{n+1:N} | x_n, y_n)$$

Under the measurement model assumption,

$$f_{Y_{n+1:N}|X_n, Y_n}(y_{n+1:N} | x_n, y_n) = f_{Y_{n+1:N}|X_n}(y_{n+1:N} | x_n),$$

so

$$f_{Y_{n:N}|X_n}(y_{n:N} | x_n) = f_{Y_n|X_n}(y_n | x_n) f_{Y_{n+1:N}|X_n}(y_{n+1:N} | x_n) \quad \blacksquare$$

Let $Z = X_{n+1}$ and apply $f_{X|Y} = \int f_{X,Z|Y} dz$:

$$\begin{aligned} f_{Y_{n+1:N}|X_n}(y_{n+1:N} | x_n) &= \int f_{Y_{n+1:N}, X_{n+1}|X_n}(y_{n+1:N}, x_{n+1} | x_n) dx_{n+1} \\ &= \int f_{Y_{n+1:N}|X_{n+1}, X_n}(y_{n+1:N} | x_{n+1}, x_n) f_{X_{n+1}|X_n}(x_{n+1} | x_n) dx_{n+1} \end{aligned}$$

By the Markov property, $Y_{n+1:N} \perp X_n | X_{n+1}$, so

$$f_{Y_{n+1:N}|X_{n+1}, X_n}(y_{n+1:N} | x_{n+1}, x_n) = f_{Y_{n+1:N}|X_{n+1}}(y_{n+1:N} | x_{n+1})$$

Hence the final derivation is

$$f_{Y_{n+1:N}|X_n}(y_{n+1:N} | x_n) = \int f_{Y_{n+1:N}|X_{n+1}}(y_{n+1:N} | x_{n+1}) f_{X_{n+1}|X_n}(x_{n+1} | x_n) dx_{n+1} \quad \blacksquare$$

5.5

Split $Y_{1:N} = (Y_{1:n-1}, Y_{n:N})$ and apply Bayes' rule, we have

$$\begin{aligned} f_{X_n|Y_{1:N}}(x_n | y_{1:N}) &= f_{X_n|Y_{1:n-1}, Y_{n:N}}(x_n | y_{1:n-1}, y_{n:N}) \\ &= \frac{f_{Y_{n:N}|X_n, Y_{1:n-1}}(y_{n:N} | x_n, y_{1:n-1}) f_{X_n|Y_{1:n-1}}(x_n | y_{1:n-1})}{f_{Y_{n:N}|Y_{1:n-1}}(y_{n:N} | y_{1:n-1})} \end{aligned}$$

In a POMP model the current state X_n renders future observations $Y_{n:N}$ independent of the past $Y_{1:n-1}$. Therefore:

$$f_{Y_{n:N}|X_n, Y_{1:n-1}}(y_{n:N} | x_n, y_{1:n-1}) = f_{Y_{n:N}|X_n}(y_{n:N} | x_n)$$

Substituting gives the final smoothing formula

$$f_{X_n|Y_{1:N}}(x_n | y_{1:N}) = \frac{f_{X_n|Y_{1:n-1}}(x_n | y_{1:n-1}) f_{Y_{n:N}|X_n}(y_{n:N} | x_n)}{f_{Y_{n:N}|Y_{1:n-1}}(y_{n:N} | y_{1:n-1})} \quad \blacksquare$$

References

Edward Ionides. *Modeling and Analysis of Time Series Data, Chapter 11: Introduction to Partially Observed Markov Process Models*. STATS 531, Winter 2026. Available at: <https://ionides.github.io/531w26/11/notes.pdf>